higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

QUANTITY SURVEYING N6

(2050026)

3 December 2019 (X-paper)
09:00–13:00

REQUIREMENTS: **Dimension paper (OE 8/12)**
Abstract paper (OE 8/10)
Billing paper (OE 8/11)

Candidates will require their own unmarked Standard System of Measuring Building Work.

Nonprogrammable calculators may be used.

This question paper consists of 5 pages and 4 addenda.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
QUANTITY SURVEYING N6
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Use only a blue or black pen.
6. Answer QUESTION 1 in the ANSWER BOOK.
7. QUESTIONS 2, 3 and 4 must be done on the appropriate sheets provided.
8. QUESTIONS 4.1 must be done on ADDENDUM D and handed in with the ANSWER BOOK.
9. ALL loose sheets must be numbered correctly and placed in sequence inside the ANSWER BOOK.
10. Candidates must strictly apply the rules and methods as stipulated in the Standard System of Measuring Building Work.
11. Write neatly and legibly.

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QUESTION 1

1.1 State the unit of measurement of the following billing items:

1.1.1 Wrought ends

1.1.2 Air bricks

1.1.3 Closing cavities in brickwork

1.1.4 Concrete in foundation cavities

1.1.5 Fair-faced brickwork

1.1.6 Timber skirtings

(6 × 1) (6)

1.2 Briefly discuss the following works when alterations to existing buildings are executed:

1.2.1 Old materials

1.2.2 Openings

(2 × 3) (6)

1.3 What are the differences between the following:

1.3.1 Specifications

1.3.2 Preambles

(2 × 4) (8)

1.4 Make a neat, section view through one stile of a single-rebated steel doorframe suitable for a 110 mm thick brick wall. (5)

[25]

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QUESTION 2

ADDENDUM A (attached) shows the front and horizontal section views of a timber cupboard made up of two hanging units with shelves at the top and a shelve and drawer unit in the centre.

Measure all the items to build these units.

SPECIFICATIONS

Carpentry and joinery	–	22 mm thick white melamine chipboard to shelves and vertical internal sides. Front edged with walnut finish edging strips	
	–	External vertical sides and front edges with a walnut finish	
	–	4 (no.) drawer units size 785 × 550 mm and 4 (no.) drawer frontals size 836 × 162 mm finished off with 2 mm thick walnut edging all round	
	–	4 (no.) doors size 496 × 2 296 mm finished off with 2 mm thick walnut edging all round	
	–	2 (no.) doors size 416 × 594 mm finished off with 2 mm thick walnut edging all round	
	–	3 mm thick hardboard backing to all units	
Ironmongery	–	190 mm long solid silver drawer handles	
	–	250 mm long solid silver door handles	
	–	350 telescope drawer runners 1 set per drawer	
	–	Concealed steel hinges 3 (no.) per cupboard door and 2 (no.) on each shelve door	
	–	Aluminium oval hanger bar including plastic brackets	
Painting	–	Two coats clear varnish to exposed timber areas	(50 ÷ 2) **[25]**

QUESTION 3

ADDENDUM B (attached) shows the top and vertical sectional views of a composite hollow block concrete slab.

Consult the applicable trades in your Standard System of Measuring Building Work and measure all the work to construct this slab.

SPECIFICATIONS

Concrete:	–	25 MPa	
Steel rod reinforcement:	–	25 MPa	
Masonry	–	Clay block size is 300 × 300 × 200 mm	
	–	Clay tile size is 150 × 150 × 20 mm	**[25]**

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QUESTION 4

ADDENDUM C (attached) shows the plan view of a dwelling and a vertical section through the roof construction. ADDENDUM D (attached) shows the measurements of the timber roof trusses and the rainwater goods.

Prepare the tender bill for the carpentry and joinery trade only.

4.1 Squaring (Answer this question on ADDENDUM D. Write your EXAMINATION NUMBER in the space provided, detach it from the question paper and hand it in with the ANSWER BOOK.) (7)

4.2 Abstracting (9)

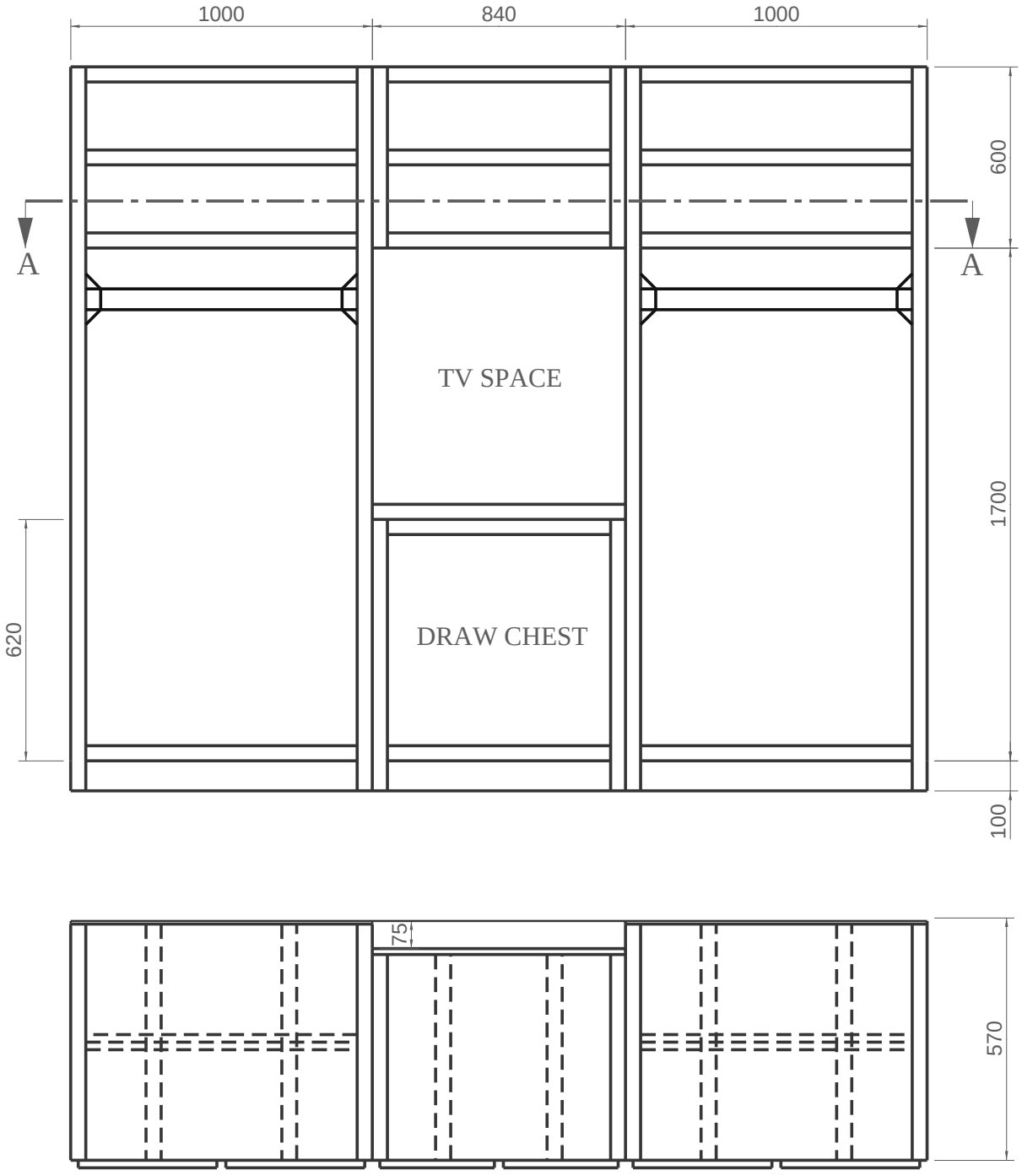
4.3 Billing (9)

[25]

TOTAL: 100

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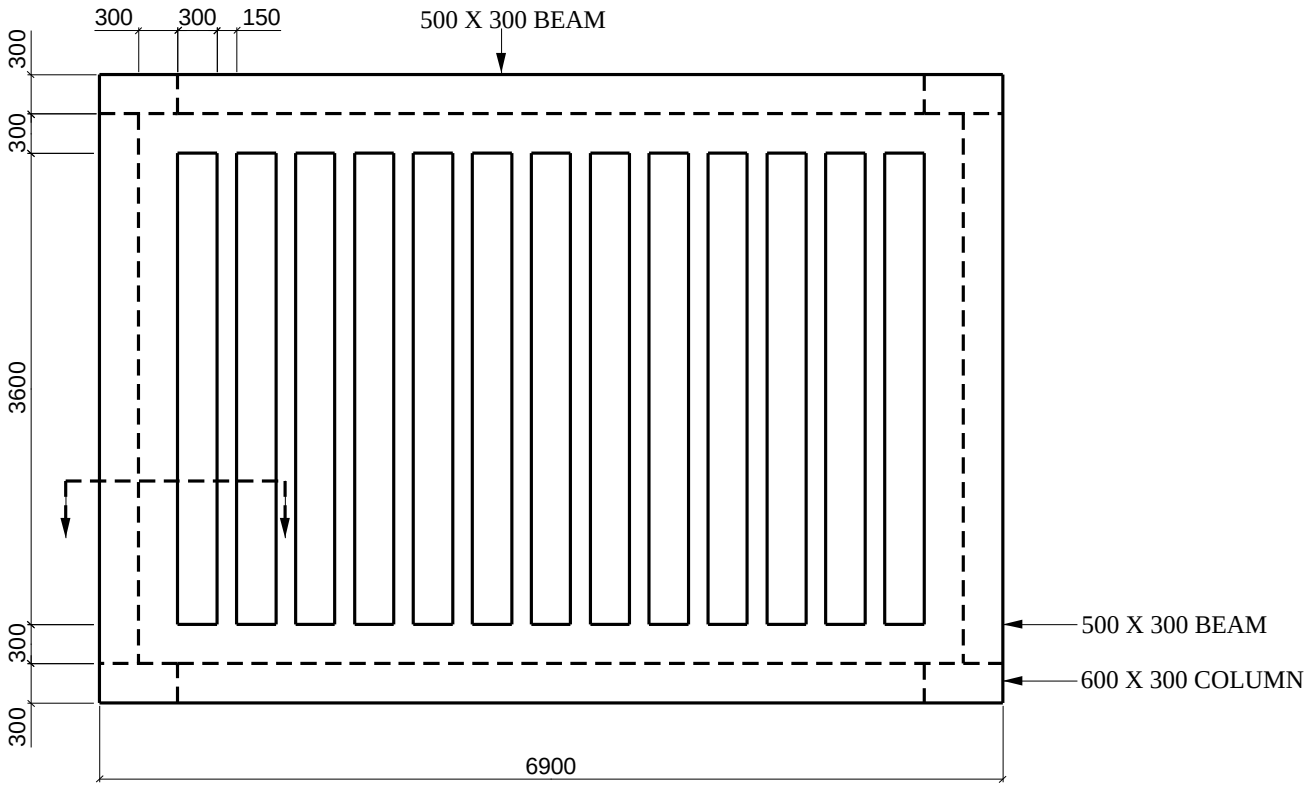
ADDENDUM A

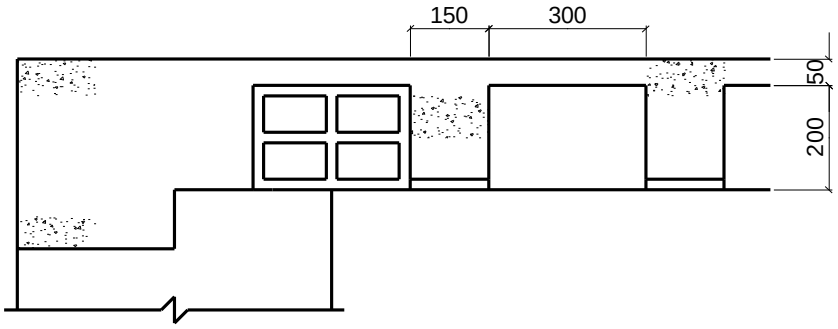


PLAN VIEW OF CUPBOARD BASE

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ADDENDUM B

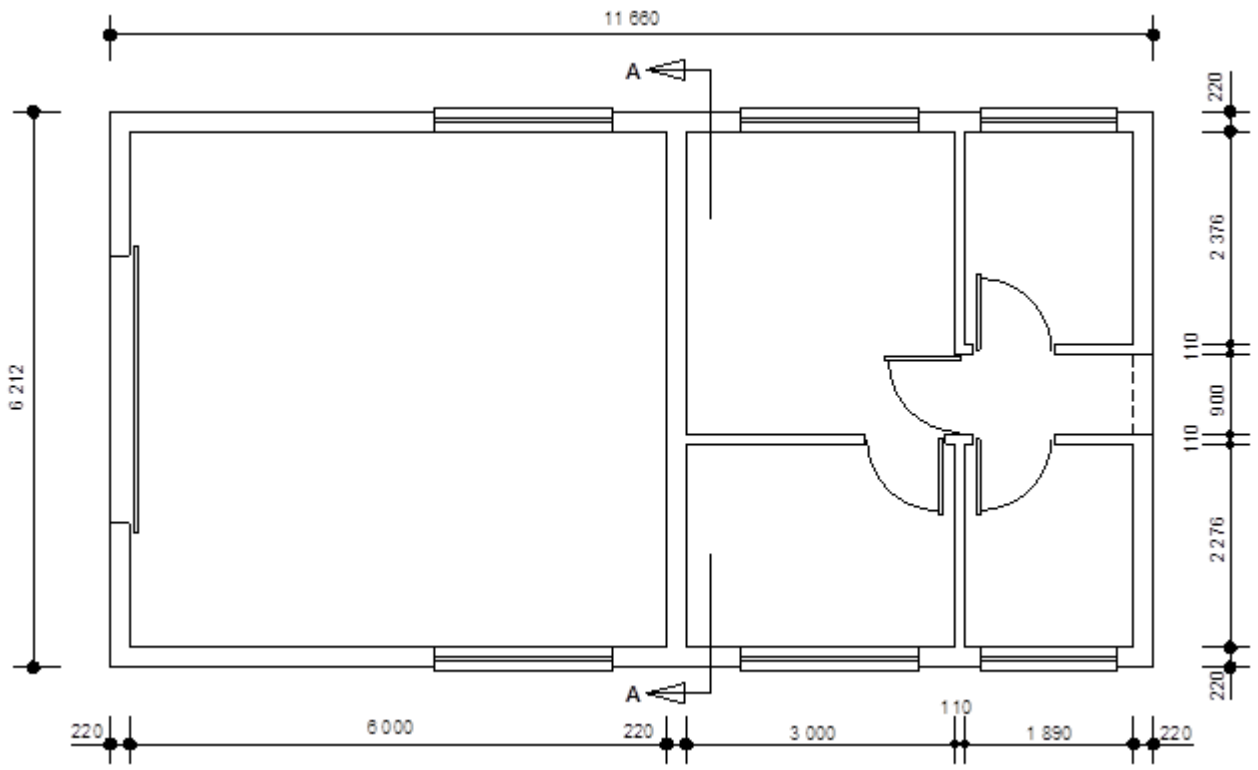


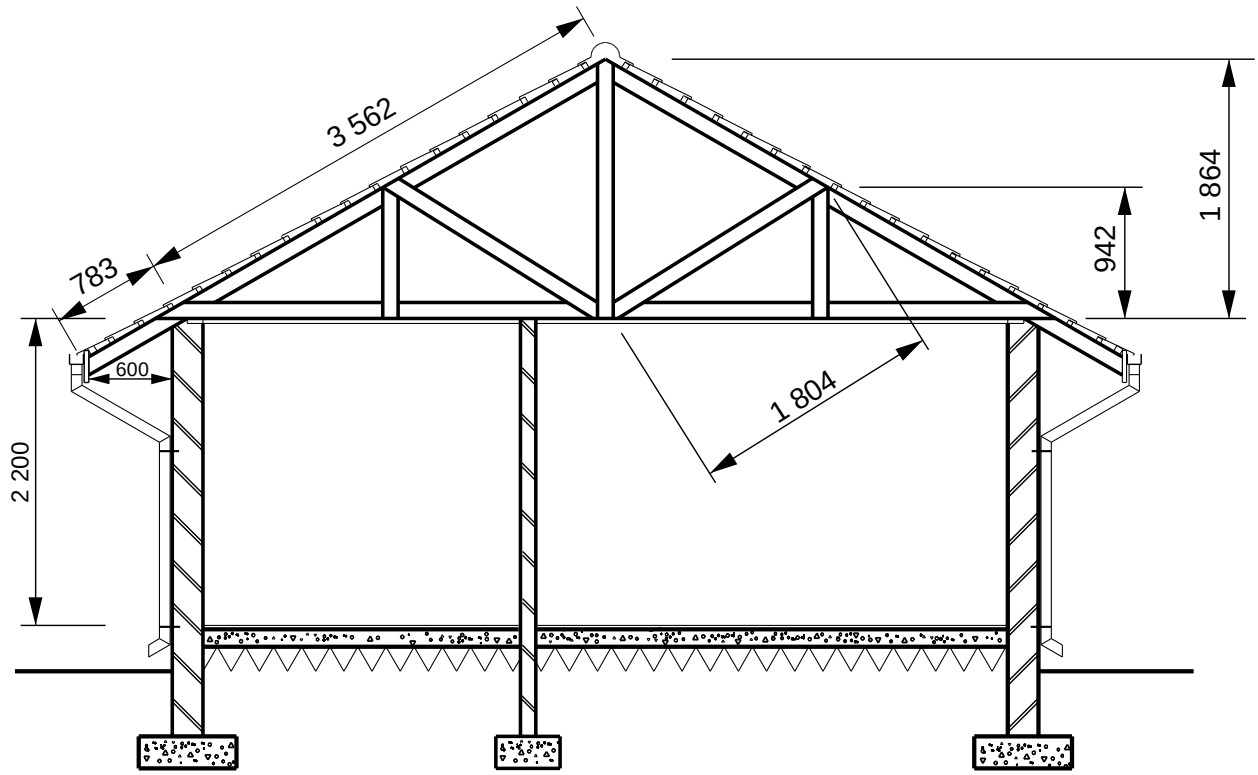


SECTION A - A

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ADDENDUM C





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ADDENDUM D **EXAMINATION NUMBER:**

2 / 11.22	38 × 114 mm S.S.A.P. w.p. 11,660 – 2(0,220) = 11,220 m and Treat tbr. w.p. with carb. oil before fixing (Super × 0,30 m = m²) 2 × 114 = 0,228 2 × 38 = 0,076 0,304 m	
11,220 0,700	=16,029 16 + 1 = 17 No. trusses The following in measurement of 17 No. full trusses fixed 2,3 m above ground level ALL x 17 No.	
17 / 17 / 2 / 1,86	38 × 114 mm S.S.A.P. tbr. n.e. 2,4 m long	
17 / 2 / 1,80	k.p.	
0,94	diag. strut q.p	
17 / 2 / 4,35		
17 / 6,21	Ditto. but exce.3,9 m n.e. 6,6 m long rafter tie bm.	
	(1)	
17 / 2		Wrought ends of tbr. rafter approx.783 mm long and Extra for vert. plumb cut to rafter ends
2 / 11,86		75 × 110 mm PVC gutters incl. brkts. screwed to tbr facia. 11,660 + 2(100) = 11,860 m
2		Extra for outlets and Extra for gutter s.e.
2 / 2,95		75 mm dia. PVC d.p. 2,200 + 0,150 + 0,600 = 2,950 m
2		Extra for swan neck and Extra for shoe
		(2)

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